

Commerce XI Mathematics & Statistics · Part 1

Unit 3: Complex Numbers

Symbols, Laws, Theorems, Results and Formulae

A quick reference based on Unit 3, textbook pages 33–43.

How to use this guide

Use	What to check
Before solving	Identify whether the question concerns powers of i , operations, square roots, quadratic equations, or cube roots of unity.
During solving	Apply the relevant identity, keep real and imaginary parts distinct, and preserve the textbook form of each expression.
Before answering	Write the result in the required form, include both square roots when required, and verify powers and signs.

Scope

Complex numbers · powers of i · conjugates · arithmetic operations · square roots · quadratic equations · cube roots of unity

1. Symbols and how to read them

Symbol	How to read it	Meaning
$i = \sqrt{-1}$	i , the imaginary unit	$i^2 = -1$
$z = a + ib$	z equals a plus i b	a is the real part; b is the imaginary part
$\text{Re}(z), \text{Im}(z)$	real part and imaginary part of z	For $z = a + ib$: $\text{Re}(z)=a$, $\text{Im}(z)=b$
$\bar{z}$	z bar; conjugate of z	If $z=a+ib$, then $\bar{z}=a-ib$
w, w^2	omega and omega squared	The two non-real cube roots of unity
$\pm$	plus or minus	Both values must be considered
$\in \mathbb{R}$	belongs to the real numbers	The stated variable is real

2. Powers of i

Power	Value	Rule
i^0	1	Any non-zero number raised to zero is 1
i^1	i	Definition
i^2	-1	Defining identity
i^3	$-i$	$i^2 \times i$
i^4	1	$i^2 \times i^2$
i^{4k}	1	k is a positive integer
i^{4k+1}	i	Remainder 1
i^{4k+2}	-1	Remainder 2
i^{4k+3}	$-i$	Remainder 3

3. Operations with complex numbers

Operation	Result
Addition	$(a+ib)+(c+id)=(a+c)+(b+d)i$
Subtraction	$(a+ib)-(c+id)=(a-c)+(b-d)i$
Multiplication	$(a+ib)(c+id)=(ac-bd)+(ad+bc)i$
Conjugate product	$(a+ib)(a-ib)=a^2+b^2$
Division	Multiply numerator and denominator by the conjugate of the denominator
Equality	$a+ib=c+id$ if and only if $a=c$ and $b=d$

4. Square roots and quadratic equations

Topic	Method or result
Square root of $x+iy$	Assume $\sqrt{x+iy}=a+ib$; then $a^2-b^2=x$ and $2ab=y$. Include both signs.
Quadratic formula	$x=(-b \pm \sqrt{b^2-4ac})/(2a)$, $a \neq 0$
Discriminant	$D=b^2-4ac$
$D < 0$ with real coefficients	The roots are a complex conjugate pair
Conjugate-root theorem	If $p+iq$ is a root of a real-coefficient quadratic, then $p-iq$ is the other root

5. Cube roots of unity

Identity	Result
Cube roots	$1, w, w^2$
Defining relation	$w^3=1$ and $w \neq 1$
Fundamental identity	$1+w+w^2=0$
Useful sum	$w+w^2=-1$
Reciprocals	$1/w=w^2$ and $1/w^2=w$
Conjugates	$\bar{w}=w^2$ and $(w^2)=w$
Power cycle	$w^{3n}=1$, $w^{3n+1}=w$, $w^{3n+2}=w^2$
General result	$1+w^n+w^{2n}=3$ if 3 divides n ; otherwise it equals 0

6. Recommended problem-solving checks

Check	What to verify
Standard form	Write the final result as $a+ib$
Powers	Reduce exponents of i modulo 4 and exponents of w modulo 3
Fractions	Rationalise with the conjugate and simplify the real denominator
Equations	Equate real parts and imaginary parts separately
Square roots	Square the proposed answer to verify the original complex number
Quadratics	Check sum and product of roots when useful